

SUPPLEMENT TO “MEASURING THE TRANSPORT OF PERTURBATION RELIABILITY RANKINGS ACROSS BIOLOGICAL CONTEXTS”

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Organization. This appendix supplies the data and source-frozen protocol (A), estimators and uncertainty decomposition (B), extended transport and independent replication evidence (C), measurement and decision robustness (D), strong comparator calibration and adaptive measurement utility, the prospective source-only boundary (E), additional robustness and diagnostic analyses (F), the external predictor context under matched information constraints (G), and the scientific regeneration path (H). It complements rather than repeats the main-text sequence: observed reordering $\rightarrow$ measurement-adjusted transport $\rightarrow$ independent replication $\rightarrow$ measurement boundary $\rightarrow$ decision consequence.

A DATA AND SOURCE-FROZEN EVALUATION PROTOCOL

The primary surface is the Frangieh Perturb-CITE-seq melanoma data set (Frangieh et al., 2021). We normalize each cell to a total of 10,000, apply $\log(1 + x)$, and retain cells with at least 700 detected genes, 1,500 counts, and at most 25% mitochondrial counts. After cell QC, the analysis uses the fixed 8,229-gene evaluation panel. Control, co-culture, and $\text{IFN}\gamma$ form the three biological contexts. A source-frozen predictor is fit at the source context and evaluated without re-training in the target context. Fig. 1B retains all 243 background rank trajectories and highlights a deterministic set of source-rank quantile exemplars for legibility; the full shortlist accounting retains all 243 labels.

The independent HepG2–Jurkat replication uses the Nadig data set (Nadig et al., 2025). Its target-sum/ $\log(1 + x)$ preprocessing is fixed before evaluation. The primary analysis requires 20 cells per condition, uses five label-disjoint folds, three frozen ensemble members, and 30 resampling seeds. These are evaluation settings, not fitted outcomes.

Supplementary Table S1: Eligibility conditions for the independent replication. The full registry, including rejected and incomplete candidates, remains a repository artifact rather than a manuscript table.

Criterion	Requirement	Nadig evidence
Shared perturbations	matched labels across contexts	2,392
Cell support	minimum 20 cells per condition	87.2% eligible
Matched controls	controls available in both contexts	4,976 / 12,013 cells
Technical comparability	same library, RNA readout, and harvest design	supported
Major context confounding	no obvious context-linked batch confound	none identified

B MEASUREMENT-ADJUSTED RELIABILITY TRANSPORT

Pairwise risk comparisons retain ties as an explicit ordering state rather than coercing them into a strict direction. Separately, exact/tolerance ties are excluded from the strict-sign Beta posterior used for the stable-direction diagnostic. A strict direction is declared stable when the 95% equal-tail Beta(1,1) posterior credible interval excludes 0.5, with at least eight non-tie observations supporting the strict comparison; otherwise the pair remains unresolved.

We report the finite-sampling floor, the measurement-adjusted discrepancy, and a conservative joint-floor stress test separately. Effect-estimate intervals resample matched perturbation labels together with the declared measurement-seed and model-member resampling units. The decision-link asso-

ciation is descriptive over 18 directed source–target surface rows; its dependence-aware sensitivity resamples three unordered context-pair blocks while preserving both directions and all metric rows. No target labels or target-derived features enter the source-frozen predictor.

For a finite set of M resampling replicates, let Z_r denote the tie-aware state for one perturbation pair in replicate r . The within-context coincidence term is estimated with the unbiased order-2 U-statistic

$$\widehat{W}_U = \frac{1}{M(M-1)} \sum_{r \neq r'} \mathbf{1}\{Z_r = Z_{r'}\}, \quad \widehat{D}_{\text{within}} = 1 - \widehat{W}_U.$$

This correction removes the positive finite-replicate bias of the plug-in coincidence estimate under a null context pair. All exact comparisons use a fixed tolerance of 10^{-8} . The strict-direction diagnostic uses a $\text{Beta}(1, 1)$ prior after excluding exact/tolerance ties; its posterior support count is the number of remaining strict observations. The metric-dependence diagnostic uses 15 prespecified discovery seeds and 15 held-out validation seeds, while the primary effect estimates retain the declared 30-seed resampling contract.

The component analysis uses the empirical pair-order law rather than the posterior tail probability used for stable-state classification. For each directed transfer and metric, both measurement replicates are retained and independent 15-seed halves form a cross-product estimator. The finite-sample report contains a tie component and a directional-bias component, compared with the canonical per-seed U-corrected D_{adj} on the same eligible universe. The population strength/flip identity is retained algebraically, but its nonlinear empirical split is not treated as unbiased. At the population level $0 \leq D_{\text{adj}} \leq 1$, but the unbiased finite-sample estimate $\widehat{D}_{\text{adj}}$ is signed and is not clipped; negative values indicate that the observed pair-state discrepancy is smaller than the finite-replicate correction. In the deterministic tie-free limit, each pair law is a point mass and D_{adj} reduces exactly to the normalized Kendall inversion fraction $(1 - \tau)/2$. This is a limit check, not a substitute for the finite-measurement estimand.

B.1 WHY THE FINITE-MEASUREMENT ESTIMAND IS NOT INTERCHANGEABLE WITH KENDALL OR GENERIC MMD

For each perturbation pair, D_{adj} compares the complete categorical laws on $\{-1, 0, +1\}$. For example, source and target laws $(0.9, 0, 0.1)$ and $(0.6, 0, 0.4)$ have the same modal strict order, so a deterministic Kendall comparison of the two mean rankings records no inversion. Nevertheless, their pair-state discrepancy is 0.09: the ordering has become substantially less reliable in the target. This is the population quantity that is absent from a point-ranking comparison. Conversely, a generic expression-space MMD can detect distributional changes that preserve every risk ordering. The exact identity with one half of squared MMD holds only for the declared delta kernel on pair states and does not turn D_{adj} into a generic expression two-sample statistic.

The U-corrected estimator is unbiased when complete measurement-seed blocks are independent and identically distributed within each context, the frozen predictor and perturbation panel do not depend on target outcomes, and the two contexts are evaluated with the declared resampling design. At the null $\pi_{a,e_1} = \pi_{a,e_2}$, its first-order influence vanishes and the order-2 statistic is degenerate. Ordinary percentile bootstrap intervals are therefore not used to decide whether transport is detectable. The calibrated procedure treats the 30 seed-level U-corrected contributions as independent units, estimates their variance directly, and uses a Student- t interval for their mean. A complete-seed block bootstrap is retained as a sensitivity analysis; the two context arms are never resampled independently.

C EXTENDED TRANSPORT EVIDENCE AND INDEPENDENT REPLICATION

The full-surface transport diagnostics contain the shared labels available to each matched context pair and are reported in the figure-source tables. Aggregate D_{adj} estimates are formed from globally merged eligible per-transfer sets; shard-local pairwise summaries are never averaged. Same-context controls are near zero (0.001–0.005), showing that the signal is not a generic artifact of the pairwise estimator.

Table S2 gives the matched-universe sensitivity of the independent Nadig replication at the two pre-specified eligibility thresholds (minimum 20 versus 40 cells per condition). The six rows cover all

Supplementary Table S3: Independent Tian raw-cell replication. Intervals are calibrated 90% Student-*t* intervals over 30 independent seed-level U-corrected contributions. All rows use 26 shared perturbations, 1,536 source-selected genes and 40 cells per condition.

Source → target	Metric	D_{adj}	90% CI	Detected
iPSC → day-7 neuron	Delta cosine	0.032	[0.005, 0.060]	yes
iPSC → day-7 neuron	Systema centroid	0.010	[−0.010, 0.030]	no
iPSC → day-7 neuron	Absolute-effect rank	0.016	[0.004, 0.028]	yes
day-7 neuron → iPSC	Delta cosine	0.049	[0.018, 0.079]	yes
day-7 neuron → iPSC	Systema centroid	0.042	[0.025, 0.060]	yes
day-7 neuron → iPSC	Absolute-effect rank	0.013	[0.001, 0.025]	yes

three metrics at each threshold; the complete two-direction full-size surface remains in the figure source table. The values are presented as measurement-adjusted components rather than as conventional prediction accuracy, keeping the replication aligned with the main claim.

For universe accounting, the Nadig registry contains 2,392 shared labels; 2,086 labels per direction meet the primary minimum-20-cell eligibility rule, and the matched sensitivity universes contain 1,255 labels at minimum 20 and 345 labels at minimum 40 after intersecting the two directions and the fixed metric panel. These counts are registry properties, not model-selected subsets.

Supplementary Table S2: Nadig matched-universe sensitivity: measurement-adjusted ordering disagreement (D_{adj} ; 90% bootstrap interval) at two minimum-support thresholds.

Universe	Metric	D_{adj} [90% CI]
min20 / 1255	Delta cosine	0.170 [0.152, 0.188]
min20 / 1255	Systema centroid	0.119 [0.092, 0.141]
min20 / 1255	Absolute-effect rank	0.108 [0.093, 0.124]
min40 / 345	Delta cosine	0.168 [0.128, 0.204]
min40 / 345	Systema centroid	0.085 [0.047, 0.124]
min40 / 345	Absolute-effect rank	0.073 [0.044, 0.103]

C.1 SECOND RAW-CELL REPLICATION: TIAN iPSC TO DAY-7 NEURON

We additionally rebuilt the full source-frozen, cell-resampled hierarchy in the Tian CRISPR screens (Tian et al., 2019). Eligibility was frozen to the 26 single-gene perturbations represented by at least 80 cells in both systems. For each source direction, the 1,536-gene evaluation panel was chosen by variance in source control cells while excluding perturbation targets. Source effects were fitted on a disjoint half of source cells; evaluation then used 40 cells per perturbation and control in each of 30 seeds with two independent measurement replicates. Target cells were absent from panel selection and prediction construction. Table S3 reports the calibrated Student-*t* intervals over per-seed U-corrected contributions. Five of six intervals exclude zero; the iPSC-to-neuron Systema row does not. This is a complete raw-cell replication of the estimand and measurement floor, not an aggregate rank-distance audit.

D MEASUREMENT AND DECISION ROBUSTNESS

Depth analyses use matched pseudoreplicates at 10, 20, 40, 80, and 160 cells per perturbation–context condition, with the full-depth estimate as reference. Detectability is the first depth whose 90% interval lower bound exceeds zero; resolution is the first depth within 10% of the full-depth estimate.

The decision link is robust over 5%, 10%, 20%, and 50% shortlist budgets, using the random-reference regret scale of the main analysis and a bounded worst-oracle scale as a secondary diagnostic. Table S4 reports the descriptive surface association and unordered-pair/source-context leave-out ranges. These rows are a compact counterpart to the main-text decision figure, not a second set of decision claims.

Supplementary Table S4: Decision-link robustness across shortlist budgets and regret normalizations. The random-reference rows use the same canonical full-depth decision surface as Fig. 4B; the bounded worst-oracle rows are a secondary source-frozen diagnostic. The last column is the leave-one-unordered-context-pair-out range.

Budget	Regret scale	ρ	Unordered-pair LOPO range
5%	random-reference normalized regret	0.769	[0.748, 0.783]
5%	worst-oracle-normalized regret	0.701	[0.657, 0.776]
10%	random-reference normalized regret	0.775	[0.755, 0.811]
10%	worst-oracle-normalized regret	0.796	[0.755, 0.818]
20%	random-reference normalized regret	0.886	[0.860, 0.916]
20%	worst-oracle-normalized regret	0.806	[0.741, 0.839]
50%	random-reference normalized regret	0.948	[0.930, 0.958]
50%	worst-oracle-normalized regret	0.889	[0.832, 0.902]

D.1 TRANSPORT DECOMPOSITION AND RANK COMPARATORS

The full-surface decomposition contains 18 directed transfer–metric rows on the matched-fixed universe (139 or 186 perturbations, hence 9,591 or 17,205 unordered pairs). Table S5 reports a finite-measurement cross-product for the tie and directional-bias components, and compares their sum with canonical per-seed U-corrected D_{adj} . The population strength/flip identity is retained algebraically, but its nonlinear empirical split is not presented as unbiased. Table S6 compares alternative rank summaries on the identical 18-row surface; these are descriptive comparator analyses and do not change the primary estimand. The depth-convergence ledger records each comparator, its same-context floor, and $|T(n) - T(\text{full})|$ for every matched depth.

Supplementary Table S5: Finite-measurement cross-product decomposition on the matched-fixed full-depth surface. The empirical report contains the tie and directional-bias components; the population strength/flip identity is not treated as an unbiased finite-sample split.

Transfer	Metric	$D_{\text{tie}}^{\text{CF}}$	$D_{\text{dir}}^{\text{CF}}$	Sum	D_{adj}	Difference
Co $\rightarrow$ Ctrl	Abs.-effect rank	0.000	0.090	0.090	0.072	0.017
Co $\rightarrow$ Ctrl	Δ cosine	0.000	0.160	0.160	0.147	0.013
Co $\rightarrow$ Ctrl	Systema	0.008	0.083	0.091	0.073	0.018
Co $\rightarrow$ IFN γ	Abs.-effect rank	0.000	0.083	0.083	0.067	0.016
Co $\rightarrow$ IFN γ	Δ cosine	0.000	0.152	0.152	0.135	0.017
Co $\rightarrow$ IFN γ	Systema	0.008	0.053	0.061	0.051	0.010
Ctrl $\rightarrow$ Co	Abs.-effect rank	0.000	0.101	0.101	0.085	0.017
Ctrl $\rightarrow$ Co	Δ cosine	0.000	0.181	0.181	0.164	0.018
Ctrl $\rightarrow$ Co	Systema	0.008	0.080	0.088	0.070	0.018
Ctrl $\rightarrow$ IFN γ	Abs.-effect rank	0.000	0.109	0.109	0.090	0.019
Ctrl $\rightarrow$ IFN γ	Δ cosine	0.000	0.156	0.156	0.134	0.022
Ctrl $\rightarrow$ IFN γ	Systema	0.008	0.067	0.075	0.060	0.015
IFN γ $\rightarrow$ Co	Abs.-effect rank	0.000	0.085	0.085	0.071	0.014
IFN γ $\rightarrow$ Co	Δ cosine	0.000	0.181	0.181	0.164	0.016
IFN γ $\rightarrow$ Co	Systema	0.008	0.050	0.058	0.044	0.014
IFN γ $\rightarrow$ Ctrl	Abs.-effect rank	0.000	0.092	0.092	0.079	0.014
IFN γ $\rightarrow$ Ctrl	Δ cosine	0.000	0.146	0.146	0.127	0.019
IFN γ $\rightarrow$ Ctrl	Systema	0.008	0.072	0.080	0.063	0.017

Supplementary Table S6: Rank-comparator benchmark on the same 18 directed transfer–metric rows. The final column is the descriptive Spearman association with normalized regret.

Comparator	n	Spearman with regret
Kendall $(1 - \tau_b)/2$	18	0.87
Spearman $(1 - \rho)/2$	18	0.84
Top-10% Jaccard distance	18	0.84
Stable-pair inversion	18	0.85
D_{adj}	18	0.78

D.2 FINITE-DEPTH DISCRIMINATION AGAINST A MATCHED NULL

The canonical summary field `measurement_floor` is the within-context term subtracted by the primary estimator; it is not itself a same-context D_{adj} shift. We therefore recomputed a matched pseudo-context null at each depth by splitting the 30 seeds into independent 15-seed halves, keeping both measurement replicates in each half. The cross-context estimate uses both held-out orientations, and the same-context null uses the two within-context half comparisons. This produces 30 macro rows and 540 surface rows across the five comparator families and six depths. Because the finite D_{adj} unit is consequential, we also compared this flattened 15-seed $\times$ 2-replicate construction with a seed-block-preserving audit that U-corrects each aligned two-replicate seed block before averaging. Across the 108 D_{adj} null rows, the current construction was negative in 108/108 rows (mean -0.00673), whereas the seed-block audit was negative in 63/108 (mean -0.000334); the maximum absolute difference was 0.0302 and rank correlation was 0.074. The constructions therefore do not estimate the same finite-sample quantity. The main-text Fig. 3C is explicitly labeled a split-half pair-law diagnostic, while the canonical primary D_{adj} remains the per-seed U-corrected summary. No negative estimate was clipped or selected away. All comparator margins were positive at every depth, but their magnitudes differ substantially; in particular, top-10 Jaccard has the largest full-depth margin whereas D_{adj} is the finite-measurement pair-state formulation, not a universally dominant rank summary. The stable-inversion comparator has lower evaluable coverage at low depth and is interpreted accordingly. In addition, we materialized a separate matched-unit audit for all five comparator families. Each aligned two-replicate seed block is collapsed to one block-level vector for the rank comparators; D_{adj} is U-corrected within each aligned block and averaged, and the stable-direction fraction uses the same 15 block-level vectors with the frozen support rule. This audit has one row per surface, depth, and comparator (540 rows) and is not substituted for the primary flattened-partition estimate. All five matched-unit margins remain positive in 108/108 rows; for D_{adj} , 3/108 matched cross estimates and 63/108 matched same-context null estimates are negative, while the margin remains positive in every row. We therefore retain the signed estimates and do not interpret the finite-sample null sign as a biological effect.

Supplementary Table S7: Finite-measurement discrimination at full depth. Each comparator is evaluated on the same 18 directed transfer-metric surfaces. The margin is the cross-context estimate minus the matched same-context pseudo-context null; the interval is a 90% unordered-context-pair bootstrap interval. These results do not rank D_{adj} as the universally best comparator.

Comparator	Cross	Same-context null	Margin [90% CI]	Recovery
Kendall distance	0.256	0.028	0.229 [0.216, 0.242]	1.00
Spearman distance	0.177	0.003	0.174 [0.157, 0.191]	1.00
Top-10 Jaccard distance	0.738	0.131	0.607 [0.594, 0.620]	1.00
Stable inversion	0.147	0.000	0.147 [0.134, 0.160]	1.00
D_{adj}	0.106	-0.005	0.111 [0.105, 0.116]	1.00

D.3 TRANSPORT-AWARE ADAPTIVE MEASUREMENT ALLOCATION

We evaluated the practical measurement-design proposal on the held-out Frangieh risk grid, starting from 10 cells per perturbation and comparing uniform, random, lowest-cell-support-first, highest-pair-state-entropy and transport-variance allocation. Acquisition scores were frozen at the pilot stage; the full-depth vectors were used only as evaluation references. The transport-variance policy reduced mean absolute error by only 1.28% on average relative to uniform allocation, with gains that varied by budget and surface. Random allocation happened to yield a larger average reduction (2.46%), but this is not an acquisition method and was not tuned or promoted. Resolution rates were identical across policies at each budget (0% through 40 cells, 5.6% at 80 cells and 33.3% at 160 cells). Thus the registered adaptive policy did not deliver a material or consistent cell saving and is retained as a negative practical-utility result rather than being moved into the main narrative. Supplementary Table S8 and the complete 450-row ledger at `artifacts/manifests/adaptive_measurement_v1/adaptive_measurement_allocation_v1.csv` record the comparison.

Supplementary Table S8: Transport-aware measurement allocation on the held-out Frangieh risk surface. Values average across the registered transfer-metric surfaces; the adaptive policy is promoted to the main text only if its error reduction and resolution gains are consistent rather than driven by one surface.

Policy	Abs. error	CI width	Detection	Resolution	Error reduction
highest pair state entropy	0.057	0.020	1.000	0.078	0.009
lowest cell support first	0.058	0.019	1.000	0.078	-0.013
random	0.056	0.019	1.000	0.078	0.025
transport variance	0.057	0.019	1.000	0.078	0.013
uniform	0.058	0.019	1.000	0.078	0.000

D.4 INCREMENTAL DECISION VALUE AND EMPIRICAL TARGET-AUDIT REPLAY

To distinguish the pooled decision association from a mean-fidelity scale effect, we fit five prespecified nested models under leave-one-unordered-context-pair-out validation. The prespecified model family is evaluated under this outer split; the nested contrasts are M1–M0, M2–M0 and M3–M1, while M4 is retained as an exploratory decomposition-feature model because it is not a superset of M3. All continuous features are standardized within the training fold. The mean-shift model gives a modest OOF MAE improvement of 0.013 with an unordered-pair block-bootstrap 90% interval $[-0.001, 0.027]$. The D_{adj} contrast beyond metric plus mean shift is -0.005 ($[-0.014, 0.003]$). These are uncertainty-aware diagnostics, not causal effects. Table S9 gives the complete OOF comparison and 2,000-draw unordered-pair block bootstrap; no headline permutation p -value is reported for three blocks.

Supplementary Table S9: Prespecified model comparison under leave-one-unordered-context-pair-out validation. ΔMAE is positive when a model improves on the stated reference; intervals are 2,000 unordered-pair block-bootstrap diagnostics. M4 is exploratory because its decomposition features are not a superset of M3.

Model	MAE	RMSE	OOF ρ	ΔMAE_{M0}	ΔMAE_{M1}	90% interval	Validation
M0 metric	0.100	0.130	0.60	0.000	–	–	ref.
M1 metric plus mean	0.087	0.119	0.66	0.013	0.000	$[-0.001, 0.027]$	LOPO (3)
M2 metric plus d adj	0.105	0.132	0.64	-0.005	–	$[-0.019, 0.008]$	LOPO (3)
M3 metric plus mean plus d adj	0.092	0.121	0.62	0.008	-0.005	$[-0.011, 0.026]$	LOPO (3)
M4 metric plus mean plus decomposition (exploratory)	0.106	0.137	0.66	-0.007	-0.019	$[-0.040, 0.027]$	LOPO (3)

The shortlist audit uses lower/upper target-risk intervals and the rules REUSE if $\text{UB}_{\text{regret}} \leq \epsilon$, RE-
VISE if $\text{LB}_{\text{regret}} > \epsilon$, and UNRESOLVED otherwise. The offline replay uses $\epsilon = 0.05$, gives every candidate a 10-cell target pilot, and spends an additional target-cell budget through the 20/40/80/160 depth ladder. Policies see only the audit pool; the disjoint reference pool is retained offline and the evaluation pool is used only for realized regret. This is an empirical decision replay, not a certification framework or a simultaneous confidence guarantee. Table S10 summarizes all 360 rows; the row-level ledger is retained in the repository.

Supplementary Table S10: Empirical target-audit replay. Every candidate receives a 10-cell pilot; the policy allocates additional target cells using only the audit pool. Values average the four decision budgets.

Policy	Mean upgraded	False REUSE	Correct REUSE	Correct REVISE
pilot only	0.0	0.000	0.333	0.028
ptl boundary	55.7	0.097	0.444	0.056
random extra depth	32.6	0.000	0.333	0.069
source uncertainty	33.0	0.000	0.333	0.125
uniform depth	144.9	0.000	0.333	0.056

D.5 REGRET-BOUND-CRITICAL ACTIVE-AUDIT REDESIGN

As a final prespecified attempt to improve the active audit, we allocated additional cells to candidates active in the regret upper bound: the highest upper-bound items inside the source shortlist and the lowest lower-bound items outside it, followed by interval width. At matched actual additional-cell cost, false reuse was eliminated relative to the legacy PTL boundary, but the policy did not exceed the source-uncertainty baseline in correct revision. The realized evaluation-pool regret is invariant under the frozen-shortlist replay by construction and is reported only as a policy-invariant reference. The kill-switch therefore demotes this policy to a negative supplementary result rather than a main-text contribution.

Supplementary Table S11: One-last active-audit redesign at matched actual additional-cell cost. The regret-bound-critical policy reduces false reuse relative to the legacy PTL boundary, but does not exceed the source-uncertainty baseline in correct revision at matched cost. Realized evaluation-pool regret is invariant under this frozen-shortlist replay by construction and is reported only as a policy-invariant reference. The prespecified kill-switch therefore demotes it to a negative supplement result.

Policy	False reuse	Correct revise	True regret	Normalized regret
Regret-bound critical	0.000	0.083	0.040	0.378
Random extra depth	0.000	0.069	0.040	0.378
Uniform depth	0.000	0.056	0.040	0.378
Source uncertainty	0.000	0.125	0.040	0.378
Legacy PTL boundary	0.097	0.056	0.040	0.378

E CAN REORDERING BE ANTICIPATED FROM SOURCE-ONLY INFORMATION?

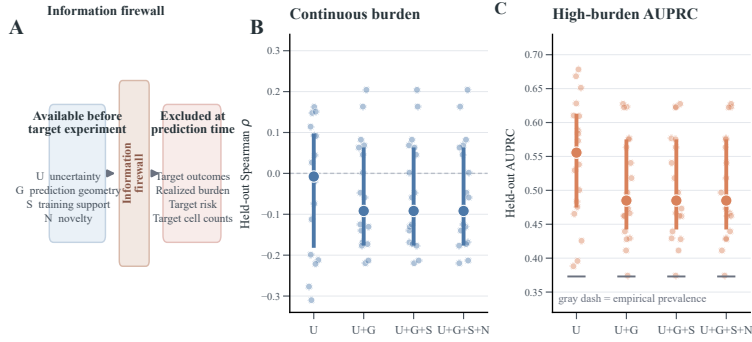
This analysis asks a narrower prospective question than the transport study: can a source-only feature ladder anticipate future reordering burden? The legal feature sets are cumulative uncertainty (U), geometry ($U + G$), support ($U + G + S$), and novelty ($U + G + S + N$). Target outcomes, target risk, and the measurement-adjusted transport value are excluded by an information firewall. The source-only task-specific usable universe contains approximately 28–44 labels; the 243-label universe belongs to the full-size shortlist surface.

The evaluation contains 18 transfer-by-metric tasks for each of continuous burden and high-burden classification. It uses ten repeats of five label-disjoint folds with nested ridge selection. Every source row records `target_feature_leakage=False`. Supplementary Fig. S1 shows held-out task-level performance directly; it intentionally does not recycle the earlier deployment-gate analysis or an unreconstructible chance baseline.

F ADDITIONAL ROBUSTNESS AND DIAGNOSTIC ANALYSES

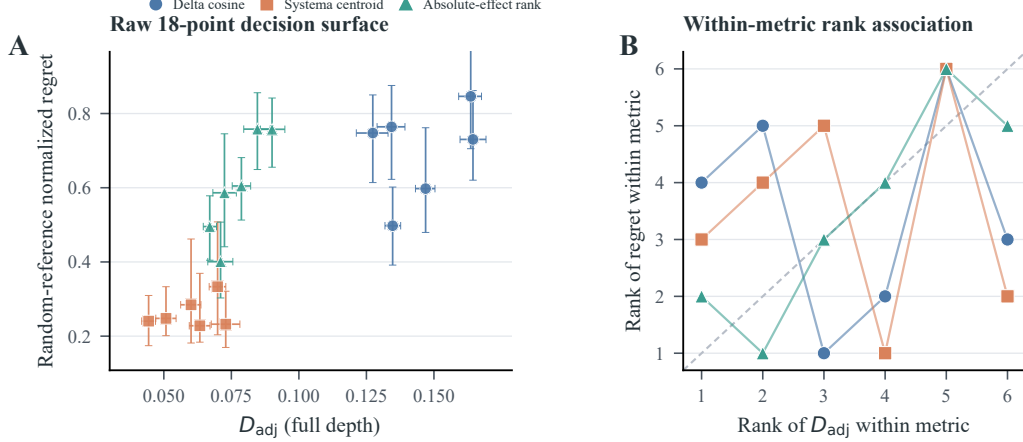
Supplementary Fig. S2 checks whether the primary decision association is explained only by the different scales of the three metrics. Panel A shows the fixed 18-point surface; Panel B ranks D_{adj} and regret within each metric before pooling. The pooled within-metric association is descriptive ($\rho = 0.26$). With only three unordered context-pair blocks, the percentile 90% block-bootstrap interval is a diagnostic, $[0.02, 0.45]$ from 2,000 draws; leave-one-pair-out estimates range from 0.00 to 0.47. The six-point metric-specific summaries are shown without a headline significance claim.

Supplementary Fig. S1 Source-only information provides limited prospective signal for reordering



Supplementary Figure S1: Source-only prospective boundary. (A) The information firewall separates source-observable features from target outcomes. (B) Held-out Spearman correlation for continuous reordering burden. (C) Held-out AUPRC for high-burden classification. Each point is one transfer-by-metric task; thick marks summarize the median and interquartile range.

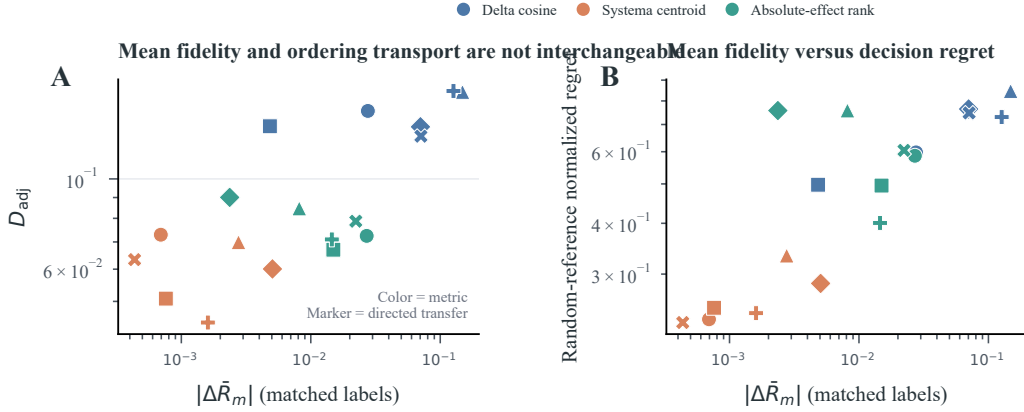
Supplementary Fig. S2 Metric-stratified decision association



Supplementary Figure S2: Metric-stratified decision association. The 18 fixed full-depth, 10% random-reference decision rows are shown in their native scale (A) and after within-metric rank transformation (B). The pooled within-metric interval is reported in the text and source table rather than overlaid on the rank points. Colors identify metrics; no target-informed feature enters the source-frozen predictor.

Supplementary Fig. S3 computes the absolute change in mean metric risk on the exact matched perturbation universe used by D_{adj} . It is a metric-matched control for the statement that a change in average fidelity and a change in reliability ordering are distinct quantities; marker shape identifies directed transfer and color identifies the metric. Across the 18 rows, the mean-risk shift has Spearman $\rho = 0.69$ with D_{adj} and $\rho = 0.77$ with normalized regret; these are descriptive matched-universe associations, not a causal decomposition.

Supplementary Fig. S3 Mean fidelity and reliability-order transport



Supplementary Figure S3: Metric-matched mean-fidelity control. Each point is one of 18 directed transfer–metric rows. The horizontal coordinate is the absolute source-to-target mean-risk shift computed on the exact fixed label universe; the vertical coordinate is D_{adj} (A) or the corresponding normalized decision regret at 10% budget (B).

Supplementary Fig. S4 validates the finite-sample estimand in synthetic data with a known null and a known alternative. The stored trial-level distributions show the positive plug-in bias under the null and the centering of the U-corrected estimator near the population identity.

G STRONG COMPARATOR CALIBRATION

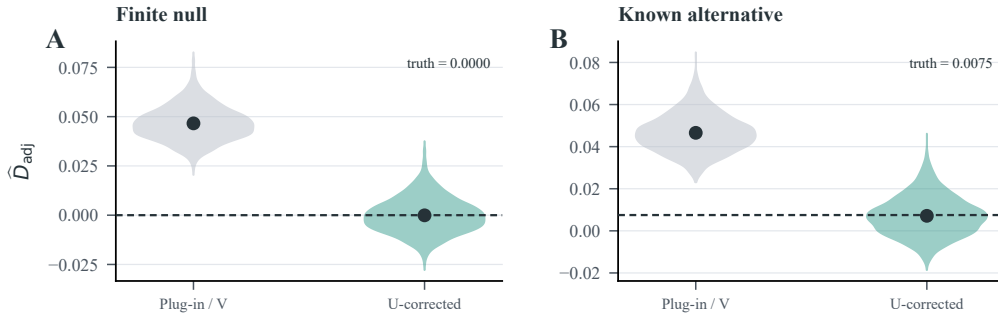
The finite-measurement claim was tested against the strongest registered traditional alternatives, not only against an uncorrected rank statistic. The benchmark contains two measurement laws, five depths, two resolution settings, two noise scales and three inversion fractions, for 120 conditions and 120 trials per condition. Every method uses the same perturbation-label resamples. The comparator set includes raw Kendall, Spearman and top-10 Jaccard, their label-bootstrap versions, matched-null-subtracted Kendall and Jaccard, stable inversion, and the U-corrected D_{adj} . The null target is zero for the measurement-adjusted quantities; the raw rank metrics are retained as descriptive sampling-disagreement comparators rather than being relabelled as transport estimands.

Raw Kendall, Spearman and Jaccard show high null false-transport rates because they compare two finite orderings without removing the same-context sampling component. Label bootstrap intervals quantify uncertainty around that comparison but do not change its target. Matched-null subtraction restores near-zero null false-transport for both Kendall and Jaccard and retains high power on the registered non-zero inversion surfaces. D_{adj} has the advantage of a single pair-state population interpretation and an explicit order-2 finite-replicate correction; the simulation does not support a claim that it universally dominates matched-null Kendall. Supplementary Table S12 reports false-transport rate, coverage and power side by side. The complete trial-level ledger is `artifacts/manifests/strong_comparator_v3/strong_comparator_v3_trials.csv`.

G.1 REAL-DATA SEMI-SYNTHETIC CALIBRATION

The known-truth simulation is complemented by a real-data semi-synthetic benchmark derived from the Frangieh raw-cell lineage. We used the canonical Control-context raw-cell pseudoreplicate risk ledger, selected the first 60 labels in the metadata-only eligible intersection, and generated two independent 12-replicate pseudo-contexts by resampling the observed replicate pool at 10, 20, 40, 80 and 160 cells. The same-context arm has a known transport null ($D_{adj} = 0$). For the controlled alternatives, a single cyclic permutation of the first 5%, 10%, 20% or 40% of the frozen label panel was applied to the target arm. This permutation was fixed before any estimates were inspected and is a response-order stress test, not a claim of a new biological replicate.

Supplementary Fig. S4 Finite-replicate validation of the ordering estimand



Supplementary Figure S4: Finite-sample validation of the ordering estimand. Trial-level sampling distributions compare the plug-in/V estimator with the order-2 U-corrected estimator under a finite-sample null and a known alternative; vertical references mark the population identity.

Supplementary Table S12: Strong comparator calibration in the registered 120-condition benchmark. Null false-transport and 90% coverage average over all zero-inversion conditions; alternative power and coverage average over the 0.1 and 0.4 inversion surfaces. MAE is to each method's own declared population target and is therefore not a universal ranking of estimators.

Method	Null FTR	Null cov.	Alt. power	Alt. cov.	Null MAE
D_{adj} (U-corrected)	0.017	0.921	1.000	1.000	0.001
Kendall	0.952	0.976	1.000	1.000	0.003
Kendall + label bootstrap	0.952	0.976	1.000	1.000	0.003
Kendall + matched null	0.003	0.965	0.997	1.000	0.003
Spearman	0.995	0.975	1.000	1.000	0.000
Top-10 Jaccard	1.000	0.783	1.000	0.987	0.029
Jaccard + label bootstrap	1.000	0.783	1.000	0.987	0.029
Jaccard + matched null	0.000	0.960	0.528	0.993	0.032
Stable inversion	0.000	0.500	0.996	0.999	0.000

The original hierarchical percentile interval resampled the two arms independently. On this discrete ledger it produced a mean null false-transport rate of 0.548 and coverage of 0.452, exposing a nonregular boundary failure rather than a biological transport signal. We therefore replaced it with the inference specified in Sec. B: complete measurement-seed blocks are the independent units and the interval is based on the empirical variance of their U-corrected contributions.

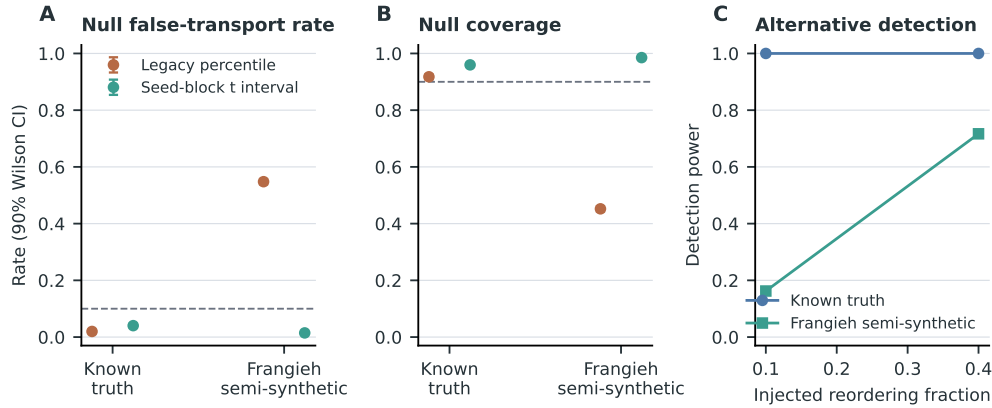
The replacement was evaluated in 2,880 known-truth null trials and 3,000 Frangieh semi-synthetic null trials. The known-truth false-transport rate was 0.040 and null coverage was 0.960; the corresponding real-data semi-synthetic values were 0.015 and 0.985. Thus the replacement is mildly conservative, particularly on the discrete empirical ledger, rather than anti-conservative at the no-transport boundary. Detection power was 1.00 at the registered 10% and 40% simulated inversion fractions; in the harder Frangieh semi-synthetic benchmark it was 0.162 and 0.717, respectively. Supplementary Fig. S5 and Table S14 compare the failed interval and calibrated replacement. The old result remains in the audit ledger but is not used for any detectability conclusion.

H EXTERNAL PREDICTOR CONTEXT UNDER MATCHED INFORMATION CONSTRAINTS

To broaden the model-family context without changing the primary estimand, we also record the completed formal-v2 external GEARS contract (Roohani et al., 2024) on three public cohorts. These canonical runs use condition-disjoint holdout, zero perturbation overlap, a dataset-native frozen 4,096-gene panel, three seeds and 20 training epochs. The two larger cohorts include held-out validation predictions; the K562 GWPS cohort has only 64 test rows and is therefore retained as a

Supplementary Table S13: Real-data semi-synthetic calibration derived from the Frangieh raw-cell lineage. The same-context null resamples the Control-context canonical risk replicates; alternatives apply one frozen cyclic ordering shift to 5–40% of the 60-label metadata-only panel. Values average over five cell depths and common hierarchical replicate-plus-label bootstrap intervals.

Method	Null FTR	Null cov.	5% power	10% power	20% power	40% power
D_{adj} (U-corrected)	0.548	0.452	0.854	0.909	0.980	0.999
Kendall	1.000	0.000	1.000	1.000	1.000	1.000
Kendall + matched null	0.038	0.961	0.150	0.254	0.500	0.771
Spearman	1.000	0.000	1.000	1.000	1.000	1.000
Top-10 Jaccard	1.000	0.132	1.000	1.000	1.000	1.000
Jaccard + matched null	0.006	0.994	0.009	0.011	0.025	0.024
Stable inversion	0.026	0.974	0.497	0.645	0.860	0.985



Supplementary Figure S5: Calibration of finite-measurement D_{adj} inference. (A) Null false-transport rate and (B) null coverage for the original percentile interval and the seed-block variance replacement; error bars are 90% Wilson intervals over trials and dashed lines mark nominal levels. (C) Detection power under controlled reordering alternatives.

test-only audit surface. Supplementary Table S15 reports the complete dataset-level summary. These results characterize conventional response prediction under external cohorts; they neither establish predictor-family invariance nor enter the Frangieh context-shift claim lock.

Supplementary Table S15: **Scope-limited external GEARS audit on public cohorts.** Each row aggregates three CUDA runs (20 epochs; seeds 0–2) under a random-condition-holdout split with zero perturbation overlap and a dataset-native frozen 4,096-gene panel. Cosine similarity is reported on non-control test cells (higher is better); MSE and MAE use the same cells (lower is better). “Validation” counts runs with held-out validation predictions. These conventional response-prediction results are not pooled with the Frangieh source-to-target reliability-transport estimand.

Dataset	Runs	Test cells	Cosine	MSE	MAE	Validation
Norman filtered	3	1,069	0.159 ± 0.058	0.01258	0.07552	3/3
Replogle RPE1	3	983	0.136 ± 0.018	0.02245	0.11218	3/3
Replogle K562 GWPS	3	64	0.069 ± 0.021	0.02716	0.12828	0/3 [†]

[†]Test-only external audit surface; no validation vector was available.

We separately executed the stricter modern-predictor bridge required for the Frangieh claim lock. The official GEARS implementation was trained for 20 epochs in five label-disjoint folds independently within each source context. Graph construction, validation and fitting used source cells only. Thirteen labels absent from the official interaction graph were removed before splitting, leaving 230 perturbations and 8,229 evaluation genes. The frozen out-of-fold vector was then passed

Supplementary Table S14: Calibration of the original percentile interval and the seed-block Student-*t* inference. Brackets are 90% Wilson intervals over null trials.

Benchmark	Inference	<i>n</i>	False transport	Null coverage
Frangieh semi-synthetic	Legacy percentile	7500	0.548 [0.538, 0.557]	0.452 [0.443, 0.462]
Frangieh semi-synthetic	Seed-block <i>t</i> interval	3000	0.015 [0.011, 0.019]	0.985 [0.981, 0.989]
Known truth	Legacy percentile	2880	0.020 [0.016, 0.025]	0.918 [0.909, 0.926]
Known truth	Seed-block <i>t</i> interval	2880	0.040 [0.035, 0.047]	0.960 [0.953, 0.965]

through the same 30-seed, two-pseudoreplicate raw-cell bridge as the other predictor families, producing the complete 18-row directed transfer-metric surface. TxPert was treated under the same information gate: the available checkpoint was trained on K562 rather than a source-matched Frangieh context, and using target basal expression would violate the frozen-source contract. It is therefore not included in the predictor-family concordance. These are deliberate contract outcomes, not missing values imputed from a different experiment; the machine-readable record is `modern_families/txpert_information_contract.json`.

An earlier one-epoch Frangieh GEARS pilot is retained only as engineering provenance and is not pooled as an additional replicate. The public-cohort GEARS audits above answer a different response-prediction question and are likewise not pooled with the strict Frangieh transport surface.

Using the same frozen panels and condition-disjoint split construction, we also completed five non-GEARS controls (45 runs total): control mean, global perturbation delta, perturbation mean delta, context-nearest-neighbour delta, and ridge regression. Together with GEARS this gives 54 comparable runs across six audited model/control families. Supplementary Table S16 reports equal-run aggregate metrics; the complete run-level CSV is retained so that the small-cohort sensitivity and model-specific variation remain auditable. Because the holdout is perturbation-disjoint, the context-*k*NN control falls back to the global mean on unseen test perturbations; we retain this row as an explicit protocol control rather than presenting it as successful neighbour transfer. These controls broaden the external benchmark context without changing the source-frozen Frangieh estimand or implying a legal second predictor family on that primary surface.

Supplementary Table S16: **Comparable external model-family audit on public cohorts.** Each entry is an equal-weight mean across nine runs (three cohorts $\times$ three seeds) on the corresponding cohort-specific frozen 4,096-gene panel and condition-disjoint holdout construction. Per-seed test-cell counts match across the audited model outputs. Cosine is higher-is-better; MSE and MAE are lower-is-better. The GEARS rows are the verified CUDA runs in Table S15; the remaining rows are deterministic or fitted baselines evaluated with the same split construction. These summaries are conventional response-prediction controls and are not pooled with the Frangieh reliability-transport claim.

Model family	Runs	Cohorts	Cosine	MSE	MAE
Ridge regression	9	3	0.233	0.01887	0.09970
Perturbation mean Δ	9	3	0.202	0.01705	0.09548
Global mean Δ	9	3	0.185	0.01737	0.09624
Context <i>k</i> NN Δ (fallback)	9	3	0.185	0.01737	0.09624
GEARS	9	3	0.122	0.02073	0.10533
Control mean	9	3	0.000	0.01823	0.09852

Equal-run means are not test-cell-weighted; the K562 GWPS cohort contributes only 64 non-control test cells per three-run cohort. Under the perturbation-disjoint holdout, the *k*NN control falls back to the global mean for unseen test perturbations; it is retained as a protocol control, not as evidence of context-neighbour transfer. Full run-level values and output paths are in `results/tables/external_condition_holdout_shared_panel_metrics.csv`.

The broader predictor-family plan was also audited against the same source-only contract. Supplementary Table S17 records the executable or semantic status of the additional families, including scGPT (Cui et al., 2024), TxPert (Wenkel et al., 2026), CPA (Lotfollahi et al., 2023), sc-Gen (Lotfollahi et al., 2019) and CellOT (Bunne et al., 2023). These rows are deliberately not treated as performance comparisons: a package, checkpoint, or context-mismatched inference run

is not evidence for the Frangieh transport claim. The corresponding machine-readable ledgers are `predictor_family_protocol.csv` and `modern_predictor_feasibility.json`.

Supplementary Table S17: **Predictor-family compatibility audit.** The audit records whether a public implementation can satisfy the manuscript’s frozen source-only prediction contract: training uses source information only, target outcomes are unseen until evaluation, and the output is a held-out vector on the declared gene panel. Only the canonical formal-v2 GEARS cohort runs enter Supplementary Tables S15–S16; the other rows are feasibility or semantic-contract records, not performance results.

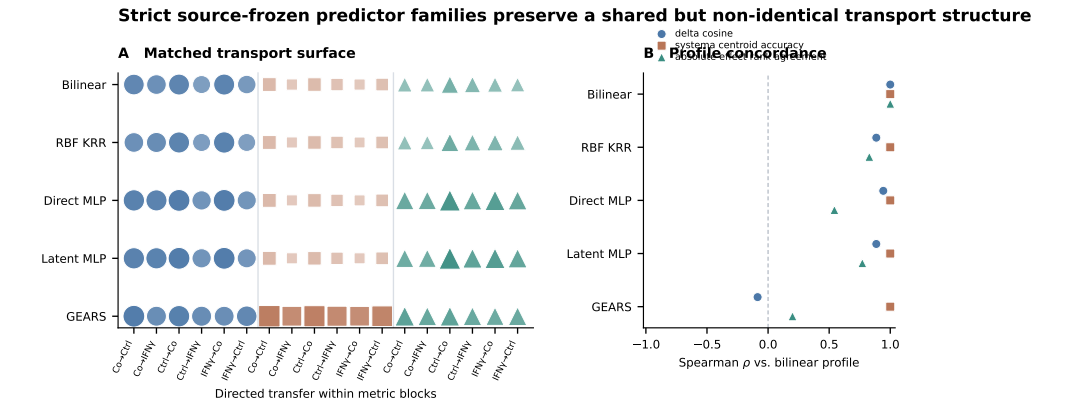
Family	Current evidence	Manuscript decision
GEARS	Formal-v2 CUDA runs completed on three public cohorts; 20 epochs; native uncertainty retained	Included only as the scope-limited external cohort audit in Supplementary Tables S15–S16; not pooled with the Frangieh estimand.
State	Official checkout and pre-processing verified; its set-transition output does not materialize the required frozen global five-fold source-only vector and target-risk bridge	Blocked from the claim lock; no score reported.
TxPert	Official cached inference is reproducible for K562, but the available checkpoint/data do not provide the frozen Nadig HepG2/Jurkat source-only vector	Blocked by context mismatch; K562 performance is not relabelled as cross-cell transfer.
scGPT	Package is installed, but import fails before model construction because the current torchtext binary cannot load (Windows error 127)	Bounded package attempt only; no model result is claimed.
biolord	Source-only adapter and pinned compatible checkpoint are not materialized in the current workspace	Compatibility pending; no score is claimed.
PerturbNet	Source-only adapter and pinned compatible checkpoint are not materialized in the current workspace	Compatibility pending; no score is claimed.
Scouter	Source-only adapter and pinned compatible checkpoint are not materialized in the current workspace	Compatibility pending; no score is claimed.
CPA-ext / scGen / CellOT	Rescue-layer implementations and weights are not information-matched to the declared source-only vector contract	Deferred to a separate context-aware rescue audit; no score is claimed.

H.1 SOURCE-ONLY PREDICTOR-FAMILY SENSITIVITY ON THE FRANGIEH SURFACE

To test whether the transport profile is tied to the primary bilinear-ridge parameterization, we materialized a source-only RBF kernel-ridge family on the same Frangieh labels, gene panel, three metrics, six directed transfers and full-size minimum-20 measurement surface. The target outcomes were loaded only after source-only prediction materialization; they were not visible during fit, penalty tuning or family selection. The initial penalty grid was extended because all 15 source-by-outer-fold choices selected its largest value. The resulting RBF sensitivity preserves the positive adjusted-

Supplementary Table S19: **Canonical source-only predictor-family transport.** Each family is evaluated on the same six directed transfers, three metrics and full-size raw-cell measurement surface. The correlation and sign columns are relative to the bilinear-ridge reference and are descriptive; they do not establish family equivalence or superiority.

Family	Mean ρ	Min point sign	Min CI sign	Mean D_{adj}	Δ mean	Rows
Bilinear ridge	1.000	1.000	1.000	0.073	0.000	6
GEARS (strict source-frozen)	0.371	1.000	1.000	0.119	0.047	6
RBF kernel ridge	0.905	1.000	1.000	0.074	0.002	6
Latent-response MLP	0.886	1.000	1.000	0.093	0.020	6
Direct MLP	0.829	1.000	1.000	0.092	0.020	6



Supplementary Figure S6: **Matched source-frozen predictor-family transport.** (A) Circle, square and triangle glyphs show D_{adj} for delta cosine, Systema centroid and absolute-effect rank; area and opacity encode magnitude. Every family uses the same six directed transfers and raw-cell bridge. (B) Within-metric profile Spearman correlation with the prespecified bilinear reference. Positive sign agreement does not imply identical transport geometry or predictor superiority.

discrepancy interval on all 18 matched rows and has descriptive D_{adj} profile Spearman $\rho = 0.983$ with the bilinear-ridge family; the within-metric profile range is 0.829–1.000 and the mean row-wise prediction-vector cosine is 0.733. Point and interval sign agreement holds on all 18 rows. This is a matched-surface sensitivity, not evidence of predictor-family equivalence, superiority, causality or broader generalization.

Supplementary Table S18: **Source-only predictor-family sensitivity on the matched Frangieh surface.** The RBF kernel-ridge family was tuned using source-only inner cross-validation; because all 15 initial source-by-outer-fold choices hit the initial largest penalty, the grid was extended to $\{30, 100, 300, 1000\}$. The comparison uses the same full-size minimum-20 label universes, frozen metrics, transfer directions and measurement-floor estimator. Source OOF fidelity is shown as the macro mean for delta cosine, Systema centroid accuracy and absolute-effect rank agreement, respectively. Concordance is descriptive and does not establish predictor-family equivalence or superiority. For the RBF family, the matched profile concordance is $\rho = 0.983$, with within-metric range 0.829–1.000 and mean row-wise prediction-vector cosine 0.733.

Family	Source-only tuning	Surfaces	Source OOF fidelity (Δ / Sys / rank)	Matched transport concordance
Bilinear ridge	Registered source-inner CV	18	0.240 / 0.499 / 0.299	Reference family
RBF kernel ridge	15/15 hit $\alpha = 10$; extended to $\alpha = 30$	18	0.257 / 0.498 / 0.310	D_{adj} profile $\rho = 0.983$; point and CI sign agreement 18/18

Supplementary Table S20: **Corrected finite-measurement calibration.** Each primary condition uses a finite-depth pair-state population as the truth for its matching estimator. Values summarize the condition-level absolute error, signed bias, RMSE, and median absolute error over the registered primary grid; the grid contains 960 conditions and 500 trials per condition. The realized pair-tie rate is reported rather than equated with the resolution setting.

Estimator	Mean MAE	Mean bias	Mean RMSE	Median MAE
D_{adj}	0.002	-0.000	0.002	0.001
Kendall distance	0.011	-0.006	0.012	0.005
Spearman distance	0.002	0.001	0.003	0.002
Top-10% Jaccard distance	0.017	0.006	0.026	0.013
Stable-pair inversion	0.010	-0.010	0.010	0.004

Mean realized pair-tie rate across conditions: 0.063. The corrected D_{adj} truth is not a latent Kendall distance; the previous v2 artifact is retained only for provenance and is not pooled here.

H.2 NEURAL SOURCE-ONLY FAMILIES AND FINITE-MEASUREMENT SIMULATION

We extended the source-only sensitivity without changing the canonical bilinear-ridge estimand. Two additional families were fit on the frozen Frangieh surface: a direct MLP and a latent-response MLP. In every outer fold, the ten-dimensional input basis and, for the latent family, the 32-dimensional response basis were fit on source training conditions only. The models used the preregistered 256–512 GELU architecture, dropout 0.1, AdamW and a source-only inner selection of weight decay. Across the three source contexts, the macro source-side OOF fidelity was 0.363 for the direct MLP and 0.366 for the latent-response MLP, compared with 0.346 for the bilinear reference and 0.355 for RBF. These are adequacy and profile diagnostics, not target-informed model selection; the target outcomes were not loaded during fitting.

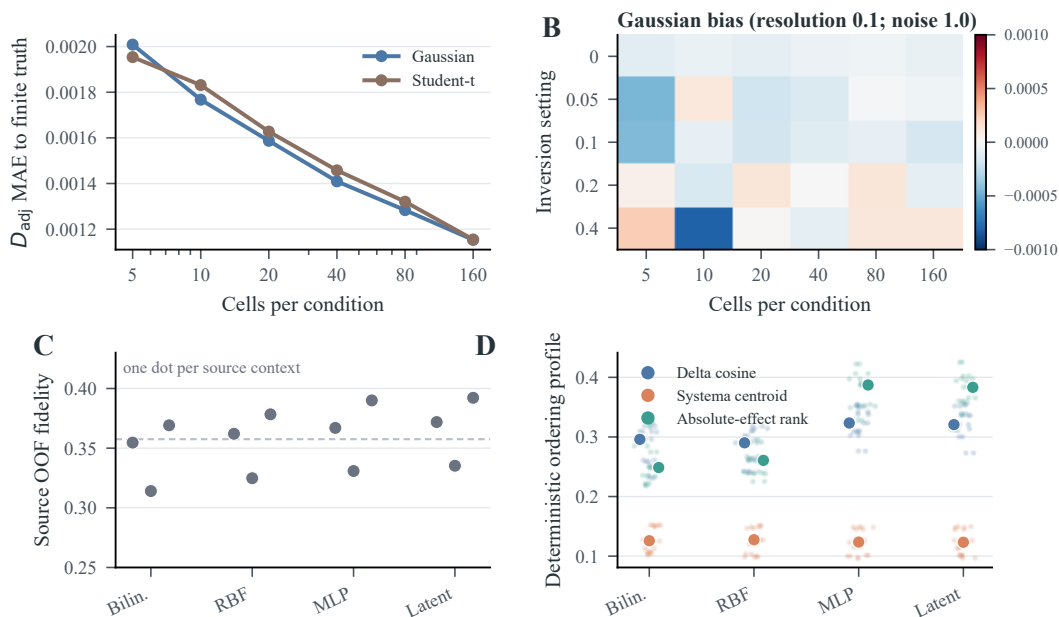
The resulting five-family profile, including strict source-frozen GEARS, is evaluated on the same frozen labels, genes, contexts and metrics. The canonical family comparison uses the same 18 directed transfer–metric rows and the same full-size raw-cell estimator; it is a sensitivity analysis, not a predictor-superiority claim. Supplementary Fig. S7 shows this profile together with the corrected finite-measurement calibration: two generative laws, 960 registered conditions (six depths, four resolution settings, four noise scales and five inversion settings), and 500 trials per condition. Here the D_{adj} truth is the finite-depth population pair-state distance at the declared resolution, while Kendall, Spearman and top-10 Jaccard use their corresponding finite-depth population-mean ordering targets. Directional states use the exact Beta posterior with the frozen minimum strict support of eight; realized pair ties are retained as a third state. The simulation is a calibration and recovery screen; it does not justify a universal superiority claim for D_{adj} or for any predictor family. Full machine-readable outputs are `predictor_families_v2/canonical_four_family_transport.csv`, `neural_family_transport_profiles.csv` and `simulation_v21/finite_measurement_primary_v21.csv`. The former `simulation_v2` artifact is retained only as provenance and is not pooled with the corrected grid.

H.3 OUTCOME-BLIND FENG MULTI-DONOR BIOLOGICAL REPLICATION AUDIT

We audited the public multi-donor human-pluripotent-stem-cell CRISPRi screen before inspecting any transport outcome (Feng et al., 2025). Eligibility was frozen from metadata support: 19 lines from 10 donors, with 135 targets supported in every line at the primary threshold of 20 cells. The response file is an aggregate target-by-line LFC table rather than raw cell-level outcomes. We therefore froze an outcome-blind 1,000-gene panel from the file schema and computed target-level RMS response risk, then compared target rankings across same-donor/different-line and different-donor pairs.

The audit is informative but not a positive replication of the full Frangieh D_{adj} hierarchy. Same-donor/different-line pairs had median Kendall distance 0.220, whereas different-donor pairs had 0.227; the corresponding Spearman distances were 0.124 and 0.131, and top-10 Jaccard distances were 0.818 and 0.700. Thus the cross-donor distance did not exceed the same-donor distance on this magnitude-rank surface. We retain this as neutral external evidence, not as a negative biological claim. A within-line cell resampling floor and hierarchical cell-level bootstrap could not be com-

Finite-measurement calibration and source-only family sensitivity



Supplementary Figure S7: **Source-only family and finite-measurement sensitivity.** (A) Mean absolute error of the corrected ordering estimate relative to the finite- depth pair-state truth across cell depth for Gaussian-heteroscedastic and Student- t measurements. (B) Bias surface for the Gaussian primary slice; the horizontal axis is cell depth and the vertical axis is the true inversion fraction. (C) Source-side OOF fidelity for each source context and predictor family. (D) Deterministic ordering profiles for the three fixed metrics across the four source-only families. The neural family and simulation analyses are descriptive and were not used to tune target transport or alter the canonical main-text estimand.

Supplementary Table S21: Outcome-blind Feng multi-donor iPSC audit. Risks are RMS response magnitudes over the frozen 1,000-gene panel, so this is a biological-replication audit rather than a full raw-cell D_{adj} hierarchy. The same-donor and different-donor rank distances are similar; no cross-donor excess is promoted.

Comparison	Pairs	Targets	Kendall d	Spearman d	Top-10 Jaccard d
Different donor	162	135	0.227	0.131	0.700
Same donor, different line	9	135	0.220	0.124	0.818

puted from the released aggregate LFC file; those analyses remain an explicit reopening requirement rather than an imputed result. Supplementary Table S21 and `artifacts/manifests/external/feng_hipsci/feng_biological_replication_pairs.csv` contain the complete pair ledger.

H.4 PROVENANCE LIMITS ON A FOUR-CONTEXT EXPANSION

We audited the exact provenance and support before adding a broader HepG2–Jurkat–K562–RPE1 transport map. HepG2 and Jurkat are both day-7 harvests, but the available Replogle K562 and RPE1 exports are day 6 and day 7, respectively, with no timepoint field in the local cell-level exports. The harvest difference is therefore inseparable from the intended context contrast. Although the four-way label intersection and minimum-support counts are large, the current 1,536-gene panel is missing 185 genes in at least one context. We consequently do not present a clean four-context biological

map; the audit is a provenance boundary, not a negative biological result. The underlying public-cohort provenance follows the Nadig atlas and the Replogle genome-scale Perturb-seq resource (Nadig et al., 2025; Replogle et al., 2022).

Supplementary Table S22: Provenance audit for a possible four-context expansion. The HepG2–Jurkat pair is a supported independent context comparison, but the available K562 and RPE1 exports differ in harvest day and cannot support a clean four-context biological map under the current contract.

Criterion	Result	Consequence
HepG2–Jurkat harvest	Both day 7	Supported independent pair
K562–RPE1 harvest	Day 6 vs day 7	Context–time confounded
Four-way shared labels	2,053	Label support is adequate
Minimum-20 support	1,688 labels	Support is adequate for profiling
Current 1,536-gene panel	1,351 present in all four	Panel cannot be reused verbatim
Four-context map	Blocked	No clean biological claim

H.5 OUTCOME-BLIND SCPERTURB3 CONTEXT AUDIT

As a separate provenance audit, we examined the local Srivatsan–Trapnell scPerturb3 file using observation and variable metadata only. The three candidate cell-line contexts (A549, MCF7 and K562) share 189 perturbations and 753 matched dose–time units per pair, with a median of at least 213 cells per matched unit. However, all 52 plates are unique to one cell line, so every cell is on a context-unique plate. This fails the preregistered no-unique-plate-confounding requirement. The source-only predictor was therefore not run on these contexts, and this audit is reported as a provenance boundary rather than as a negative biological result. The complete outcome-blind record is `artifacts/manifests/sciplex3_context_replication_audit.v2.json`; the corresponding evidence and reopening requirements are listed in `EXTERNAL_DATA_NEEDED.V2.md`.

I REPRODUCIBILITY AND SCIENTIFIC REGENERATION

The scientific regeneration path starts from the canonical manifest tables, fixed splits and resampling settings, and rebuilds the four main figures, Supplementary Figs. S1–S5, and the compact supplementary tables. The V3 comparator, Feng and adaptive ledgers are generated from their corresponding scripts and are linked to the tables above; they do not alter the canonical main figures. The anonymous repository documents the executable commands and data lineage. Large raw H5AD/NPZ inputs remain external to the manuscript tree; the reviewable CSV/JSON products expose the values used in figures and tables.

Code and data availability. The anonymized core release is available at the anonymous core branch and contains the source code, repository-relative configurations, manuscript sources, publication figures, and compact reviewable data products. Raw single-cell matrices are not redistributed. The Frangieh RNA screen and the Nadig HepG2–Jurkat screens are available from the public scPerturb v1.4 Zenodo record (Zenodo); the repository README documents the relevant file names and regeneration commands.

The scope-limited external GEARS audit and its matched baseline controls are summarized in Supplementary Tables S15 and S16; the broader predictor-family compatibility ledger is summarized in Table S17. Broader condition-holdout analyses, old failure-anatomy graphics, and the earlier broad source-only deployment gate remain repository-only. Runtime compatibility notes for predictor families that do not satisfy the frozen source-only contract are likewise retained as an engineering ledger, not presented as primary scientific evidence.

REFERENCES

Charlotte Bunne, Stefan G. Stark, Gabriele Gut, Jacobo Sarabia del Castillo, Mitch Levesque, Kjong-Van Lehmann, Lucas Pelkmans, Andreas Krause, and Gunnar Raetsch. Learning single-

- cell perturbation responses using neural optimal transport. *Nature Methods*, 20(11):1759–1768, 2023. doi: 10.1038/s41592-023-01969-x.
- Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.
- Claudia Feng, Elin Madli Peets, Yan Zhou, et al. A genome-scale single-cell CRISPRi map of trans gene regulation across human pluripotent stem cell lines. *Cell Genomics*, 5, 2025. doi: 10.1016/j.xgen.2025.101076. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC12903452/>. Public multi-line, multi-donor targeted-screen resource accessed for an outcome-blind external replication audit.
- Chris J. Frangieh, Johannes C. Melms, Pratiksha I. Thakore, Kathryn R. Geiger-Schuller, Patricia Ho, Adrienne M. Luoma, Brian Cleary, Livnat Jerby-Arnon, Shruti Malu, Michael S. Cuoco, Maryann Zhao, Casey R. Ager, Meri Rogava, Lila Hovey, Asaf Rotem, Chantale Bernatchez, Kai W. Wucherpennig, Bruce E. Johnson, Orit Rozenblatt-Rosen, Dirk Schadendorf, Aviv Regev, and Benjamin Izar. Multimodal pooled Perturb-CITE-seq screens in patient models define mechanisms of cancer immune evasion. *Nature Genetics*, 53:332–341, 2021. doi: 10.1038/s41588-021-00779-1.
- Mohammad Lotfollahi, F. Alexander Wolf, and Fabian J. Theis. scGen predicts single-cell perturbation responses. *Nature Methods*, 16:715–721, 2019. doi: 10.1038/s41592-019-0494-8.
- Mohammad Lotfollahi, Anna Klimovskaia Susmelj, Carlo De Donno, Leon Hetzel, Yuge Ji, Ignacio L. Ibarra, Sanjay R. Srivatsan, Mohsen Naghipourfar, Riza M. Daza, Beth Martin, Jay Shendure, Jose L. McFaline-Figueroa, Pierre Boyeau, F. Alexander Wolf, Nafissa Yakubova, Stephan G’unnemann, Cole Trapnell, David Lopez-Paz, and Fabian J. Theis. Predicting cellular responses to complex perturbations in high-throughput screens. *Molecular Systems Biology*, 19(6), 2023. doi: 10.15252/msb.202211517.
- Ajay Nadig, Joseph M. Replogle, Angela N. Pogson, Mukundh Murthy, Steven A. McCarroll, Jonathan S. Weissman, Elise B. Robinson, and Luke J. O’Connor. Transcriptome-wide analysis of differential expression in perturbation atlases. *Nature Genetics*, 57:1228–1237, 2025. doi: 10.1038/s41588-025-02169-3.
- Joseph M. Replogle, Reuben A. Saunders, Angela N. Pogson, Jeffrey A. Hussmann, Alexander Lenail, Alina Guna, Lauren Mascibroda, Eric J. Wagner, Karen Adelman, Gila Lithwick-Yanai, Nika Iremadze, Florian Oberstrass, Doron Lipson, Jessica L. Bonnar, Marco Jost, Thomas M. Norman, and Jonathan S. Weissman. Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575.e28, 2022. doi: 10.1016/j.cell.2022.05.013.
- Yusuf Roohani, Kexin Huang, and Jure Leskovec. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nature Biotechnology*, 42(6):927–935, 2024. doi: 10.1038/s41587-023-01905-6.
- Ruilin Tian, Mariam A. Gachechiladze, Connor H. Ludwig, Matthew T. Laurie, Jason Y. Hong, Diane Nathaniel, Anika V. Prabhu, Michael S. Fernandopulle, Rajan Patel, Mehrnoosh Abshari, Michael E. Ward, and Martin Kampmann. Crispr interference-based platform for multimodal genetic screens in human ipsc-derived neurons. *Neuron*, 104(2):239–255.e12, 2019. doi: 10.1016/j.neuron.2019.07.014.
- Frederik Wenkel, Wilson Tu, Cassandra Masschelein, Hamed Shirzad, Liam Hodgson, Ihab Bendidi, Cian Eastwood, Shawn T. Whitfield, Craig Russell, Yassir El Mesbahi, Jiarui Ding, Marta M. Fay, Berton Earnshaw, Emmanuel Noutahi, and Alisandra K. Denton. TxPert: using multiple knowledge graphs for prediction of transcriptomic perturbation effects. *Nature Biotechnology*, 2026. doi: 10.1038/s41587-026-03113-4.